Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS with strengths in C++, Python, algorithms, systems reasoning, and large-scale experimentation, with research and software projects spanning distributed workflows, simulation, debugging, and production-oriented engineering.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and model-training pipelines** for autonomous-driving planning, processing 100K+ trajectory logs while improving long-horizon planning by **18%** through data preparation, evaluation tooling, debugging, and systems analysis.
- Architected a **spatial-temporal encoder** with multi-head attention over map and agent context, improving closed-loop planning stability by **15%** while strengthening large-scale experimentation, analytical problem solving, and model debugging.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*
- Analyzed **consensus mechanisms and cryptographic designs** for quantum-resistant blockchain systems, developing a novel algorithm that reduced centralized-control and quantum-risk exposure by **25%** in benchmark simulations for distributed protocols.
- Contributed to a **peer-reviewed technical paper** and presented the work at IISc Bengaluru, strengthening distributed-systems reasoning, technical communication, research rigor, and systems thinking in collaborative engineering settings.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform APIs and ETL services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed data integrations, transformation workflows, and dependable release coordination.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** while strengthening defect isolation, repeatable testing, and cross-platform delivery across production systems.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and data workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, schema updates, and maintainable release pipelines across cloud-hosted systems and internal operations.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Flask backend services** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and model-backed evaluation pipelines while supporting debugging and stable API behavior under load.
- Built **concurrent batch-processing workflows** for uploaded files, reducing feedback turnaround by **60%** while improving issue isolation, testing confidence, performance reasoning, and throughput stability across production-facing application components.

Skills

Languages: C++, Python, C, Java, JavaScript, SQL, Bash
Systems: Linux, multi-threaded execution, data structures, algorithms, OO design, distributed workflows, real-time reasoning
Development: Git/GitHub, Jenkins, build pipelines, debugging, unit testing, integration testing, CI/CD, scripting, version control, build tools
Related Areas: REST APIs, SQL/NoSQL databases, networking fundamentals, cloud services, cross-platform development, Python scripting, TCP/IP basics

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **FCFS, SJF, SRT, RR, priority scheduling, and MLQ** with C++ logic and dynamic visualizations, reducing debugging time by **20%** across 1K+ simulated process combinations and timing analysis.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Worked across a **Java and TypeScript codebase** with Docker, Maven, Node.js, and Yarn to support cross-platform database workflows on Windows, macOS, and Linux environments for developer tooling.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a **scalable fintech platform** supporting 50K+ users and 1M+ transactions with real-time synchronization, secure integrations, CI/CD-backed deployment workflows, and operational reliability across distributed application components overall.